



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 13

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Program:

```
#include <stdio.h>

int max(int a, int b)
{
    return (a > b) ? a : b;
}

int maxCrossingSum(int a[], int left, int mid, int right)
{
    int sum = 0, leftSum = -999999, rightSum = -999999;
    int i;
    for (i = mid; i >= left; i--)
    {
        sum += a[i];
        leftSum = max(leftSum, sum);
    }
    sum = 0;

    for (i = mid + 1; i <= right; i++)
    {
        sum += a[i];
```

```

    rightSum = max(rightSum, sum);
}
return leftSum + rightSum;
}
int divideAndConquer(int a[], int left, int right)
{
    int mid, leftSum, rightSum, crossSum;
    if (left == right)
        return a[left];
    mid = (left + right) / 2;
    leftSum = divideAndConquer(a, left, mid);
    rightSum = divideAndConquer(a, mid + 1, right);
    crossSum = maxCrossingSum(a, left, mid, right);
    return max(max(leftSum, rightSum), crossSum);
}
int kadane(int a[], int n)
{
    int current = a[0], maximum = a[0];
    int i;
    for (i = 1; i < n; i++)
    {
        current = max(a[i], current + a[i]);
        maximum = max(maximum, current);
    }
    return maximum;
}

int main()
{

```

```

    int a[] = {-2, -3, 4, -1, -2, 1, 5, -3};

    int n = 8;

    printf("Name: Dharshana Sivakumar\n");

    printf("Maximum Subarray Sum\n\n");

    printf("Kadane's Algorithm: %d\n", kadane(a, n));

    printf("Divide and Conquer: %d\n", divideAndConquer(a, 0, n - 1));


    return 0;

}

```

Output:

main.c	Run ▶	Output
<pre> 1 #include <stdio.h> 2 3 int max(int a, int b) 4 { 5 return (a > b) ? a : b; 6 } 7 8 int maxCrossingSum(int a[], int left, int mid, int right) 9 { 10 int sum = 0, leftSum = -999999, rightSum = -999999; 11 int i; 12 13 for (i = mid; i >= left; i--) 14 { 15 sum += a[i]; 16 leftSum = max(leftSum, sum); 17 } 18 19 sum = 0; 20 21 for (i = mid + 1; i <= right; i++) 22 { 23 sum += a[i]; 24 rightSum = max(rightSum, sum); 25 } 26 27 return leftSum + rightSum; 28 } 29 30 int divideAndConquer(int a[], int left, int right) 31 { 32 int mid, leftSum, rightSum, crossSum; 33 </pre>		<pre> Name: Dharshana Sivakumar Maximum Subarray Sum Kadane's Algorithm: 7 Divide and Conquer: 7 </pre>